

Machine Learning

1. What is Bias and Variance?

Bias

- Error caused by overly simple assumptions.
- High bias → model cannot learn patterns.
- Causes **underfitting**.

Variance

- Error caused by being too sensitive to training data.
- High variance → model memorizes training data.
- Causes **overfitting**.

Example

- Linear model for complex data → High bias.
 - Very deep neural network on small dataset → High variance.
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2. What is Underfitting?

Underfitting occurs when the model is too simple to capture the relationship in the data.

Characteristics

- Low training accuracy
- Low testing accuracy
- High bias

Solutions

- Increase model complexity
 - Add more useful features
 - Train longer
 - Reduce regularization
-

3. What is Cross-Validation?

Cross-validation evaluates a model by repeatedly splitting the dataset into training and validation sets.

Most common:

K-Fold Cross Validation

Example:

K = 5

- Split data into 5 parts
- Train on 4
- Validate on 1
- Repeat 5 times
- Average the accuracy

Advantages

- Better estimate of model performance
 - Reduces overfitting risk
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4. What is a Confusion Matrix?

A confusion matrix measures classification performance.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

From this we calculate

- Accuracy
- Precision
- Recall
- F1-score

Here are short, viva-friendly definitions:

- **Accuracy:** Measures the overall percentage of correct predictions made by the model.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** Measures how many of the predicted positive cases are actually positive.

$$\left[\text{Precision} = \frac{TP}{TP + FP} \right]$$

- **Recall (Sensitivity):** Measures how many of the actual positive cases are correctly identified by the model.

$$\left[\text{Recall} = \frac{TP}{TP + FN} \right]$$

- **F1-Score:** The harmonic mean of Precision and Recall, providing a balance between the two.

$$\left[\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \right]$$

Easy way to remember (for viva)

- **Accuracy:** Overall correctness of the model.
- **Precision:** When the model says "Yes," how often is it correct?
- **Recall:** Of all actual "Yes" cases, how many did the model find?
- **F1-Score:** A single score that balances Precision and Recall.

One-line example

Suppose an AI model detects diseased mangoes:

- **Accuracy:** Out of all mangoes, how many were classified correctly?
- **Precision:** Out of all mangoes predicted as diseased, how many were actually diseased?
- **Recall:** Out of all actually diseased mangoes, how many did the model detect?
- **F1-Score:** A balanced measure of Precision and Recall, especially useful when classes are imbalanced.

5. What is Feature Engineering?

Feature engineering is creating or selecting useful input features to improve model performance.

Examples:

- Age from Date of Birth
- BMI from height and weight
- One-hot encoding
- Normalization

Good features usually improve accuracy.

6. Normalization vs Standardization

Normalization

Scales data between 0 and 1

Formula:

$$\left[\frac{x - \min}{\max - \min} \right]$$

Best for:

- Neural Networks
 - Image processing
-

Standardization

Transforms data to have

- Mean = 0
- Standard deviation = 1

Formula

1. Calculate mean
2. Calculate standard deviation
3. Calculate Z-Score

$$\text{Z-Score} = \left[\frac{x - \mu}{\sigma} \right]$$

Best for

- Logistic Regression
 - SVM
 - PCA
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7. Difference between Batch Learning and Online Learning

Batch Learning

- Uses entire dataset at once
- Training happens offline
- Retrain from scratch

Examples

- Image classification
- Offline ML

Online Learning

- Learns continuously
- Updates after each new sample
- Suitable for streaming data

Examples

- Stock prediction
 - Recommendation systems
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8. Why split data into Training, Validation and Testing?

Training set

Used to train the model.

Validation set

Used to tune hyperparameters.

Test set

Used only for final evaluation.

This prevents overfitting and gives an unbiased estimate.

Typical split:

70%

15%

15%

or

80%

10%

10%

9. Which optimizer did you use in your project and why?

Answer (your thesis):

I used the **Adam optimizer**.

Reasons:

- Fast convergence
- Adaptive learning rate
- Works well for deep learning
- Less parameter tuning
- Stable training

Adam combines Momentum and RMSProp.

10. Training accuracy 100% but poor testing accuracy

Problem:

Overfitting

Solutions:

- Collect more data
 - Data augmentation
 - Early stopping
 - Dropout
 - L2 Regularization
 - Simpler model
 - Cross-validation
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Artificial Intelligence / Prolog

11. Why Prolog instead of C/C++?

Because Prolog is designed for Artificial Intelligence.

Advantages:

- Logic programming
- Automatic inference
- Pattern matching
- Backtracking
- Knowledge representation

C/C++ requires manually implementing these.

12. What is a Fact?

A fact represents information that is always true.

Example

```
parent(john,mary).
```

13. What is a Rule?

A rule defines logical relationships.

Example

```
grandparent(X,Y):-  
parent(X,Z),  
parent(Z,Y).
```

14. What is a Query?

A query asks Prolog to find answers.

Example

```
?- parent(john,mary).
```

Output

```
true.
```

15. What is Backtracking?

If Prolog cannot satisfy a condition, it automatically goes back and tries another possible solution.

This searching process is called backtracking.

16. How does Prolog perform inference?

Using:

- Facts
- Rules

- Queries
- Unification
- Backtracking

It searches through rules until it finds a solution.

17. How did you implement BFS?

Using

Queue

Steps

1. Start node
 2. Push into queue
 3. Visit node
 4. Push neighbors
 5. Repeat until queue empty
-

18. How did you implement DFS?

Using

Stack

or

Recursion

Steps

1. Visit node
 2. Visit first child
 3. Continue deeply
 4. Backtrack
-

19. Which data structure is used in BFS?

Queue

FIFO

20. Which data structure is used in DFS?

Stack

or recursion.

LIFO

21. Why is Genetic Algorithm suitable for optimization?

Because it searches large solution spaces efficiently using:

- Selection
- Crossover
- Mutation

It can avoid local optimum and find near-optimal solutions.

Data Mining

22. What is Knowledge Discovery?

Knowledge Discovery (KDD) is the process of discovering useful patterns from large datasets.

Steps

- Cleaning
 - Integration
 - Selection
 - Transformation
 - Data Mining
 - Evaluation
 - Presentation
-

23. What is Data Cleaning?

Removing:

- Missing values
- Duplicate records
- Incorrect values
- Noise

Improves data quality.

24. What is Data Transformation?

Converting data into suitable format.

Examples

- Normalization
 - Aggregation
 - Encoding
-

25. What is Data Reduction?

Reducing dataset size while preserving useful information.

Methods

- Sampling
 - PCA
 - Feature selection
-

26. What is Noise in Data?

Random or incorrect data.

Examples

- Typing mistakes
 - Sensor errors
 - Outliers
-

27. Structured vs Unstructured Data

Structured

- Tables
- SQL
- Fixed schema

Examples

Excel

Database

Unstructured

- Images
- Videos
- Audio
- Text

No fixed format.

28. What is Frequent Itemset?

Items that appear together frequently.

Example

Bread + Milk

Used in Market Basket Analysis.

29. Sequential Pattern Mining

Finds patterns based on sequence.

Example

Customer buys

Phone

↓

Case

↓

Earphone

30. Why OLAP- Online analytical Procedure?

Online Analytical Processing (OLAP) is a computing approach that allows users to extract and analyze massive volumes of historical data from multiple dimensions.

It powers business intelligence (BI) by enabling fast, complex, self-service queries for tasks like forecasting, financial reporting, and trend analysis

OLAP provides

- Fast analysis
- Multidimensional view
- Better decision making

Used in business intelligence.

31. Why Hadoop is Fault Tolerant?

Because HDFS stores multiple copies (replicas) of data.

If one node fails,

Another replica is used.

32. SQL vs NoSQL

SQL

- Relational
- Fixed schema
- ACID
- Vertical scaling

NoSQL

- Flexible schema
- Horizontal scaling
- Large data

- JSON documents
-

33. Why MongoDB is called Document Database?

Because it stores data as JSON-like BSON documents instead of tables.

34. What is a Partitioner?

Partitioner decides which reducer receives each key-value pair.

This balances workload among reducers.

35. Hadoop vs Spark

Hadoop

- Disk-based
- Slower
- Large batch processing

Spark

- In-memory
- Faster
- Better for iterative machine learning

Here are short, viva-friendly definitions:

Hadoop

Hadoop is an **open-source framework** that stores and processes **large datasets across multiple computers** using distributed storage (HDFS) and parallel processing (MapReduce).

One-line viva answer:

Hadoop is a framework for distributed storage and parallel processing of big data.

Spark

Apache Spark is an **open-source big data processing framework** that performs **fast, in-memory data processing** for analytics and machine learning.

One-line viva answer:

Spark is a fast, in-memory distributed computing framework for big data processing.

Difference (Very Common Viva Question)

Hadoop	Spark
Disk-based processing	In-memory processing
Slower	Faster
Uses MapReduce	Uses DAG execution engine
Good for batch processing	Good for batch, streaming, ML, and analytics

Easy to remember:

- **Hadoop = Stores + Processes Big Data**
 - **Spark = Processes Big Data Faster 🚀**
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Software Quality Assurance (SQA)

36. What is Software Quality?

Software quality means software satisfies user requirements, is reliable, efficient, secure, and free from defects.

37. What is Quality Assurance?

QA is a process-oriented activity that prevents defects by improving development processes.

38. QA vs QC

QA

- Prevents defects
- Process-focused

QC

- Finds defects
 - Product-focused
-

39. Verification vs Validation

Verification

Verification checks whether the software is built **according to the specifications**.

Easy to remember:

"Are we building the product right?"

Validation

Validation checks whether the software **meets the user's needs and requirements**.

Easy to remember:

"Are we building the right product?"

Viva Difference

Verification	Validation
Checks the development process	Checks the final product
No software execution required	Requires software execution
Done before or during development	Done after development/testing

One-line Trick

- **Verification = Build it right. Perform by developer**
 - **Validation = Build the right product. Perform by tester or user**
-

40. White Box vs Black Box Testing

White Box

- Internal code tested
- Programmer knows code

Black Box

- Test functionality
 - No knowledge of code
-

41. Alpha vs Beta Testing

Alpha

- Inside company
- Developers/testers

Beta

- Real users
 - Before official release
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42. What is a Test Case?

A document containing

- Test ID
 - Objective

 - Inputs data
 - Steps to test
 - Expected output
 - Actual output
 - Test status
-

43. What is a Test Scenario?

A high-level description of what should be tested.

Example

"Login functionality."

44. What is Inspection?

Formal review of documents or code to find defects without executing the software.

45. What is Walkthrough?

An informal review where the author explains the software or document to peers for feedback.

46. What is Configuration Management?

Managing software versions, changes, and releases systematically.

47. What is Git?

Git is a distributed version control system used to track changes and support collaboration.

48. What is ISO 9001?

An international standard for Quality Management Systems (QMS) that helps organizations consistently meet customer and regulatory requirements.

49. Five CMM Levels

CMM (Capability Maturity Model)

Definition (Short):

CMM (Capability Maturity Model) is a framework used to **assess and improve an organization's software development process**.

Purpose

- Improves software quality
- Makes processes more organized and predictable

- Reduces defects

Five Maturity Levels

1. **Initial** – Process is unpredictable and ad hoc.
2. **Managed (Repeatable)** – Basic project management practices are established.
3. **Defined** – Processes are documented and standardized across the organization.
4. **Quantitatively Managed** – Processes are measured and controlled using metrics.
5. **Optimizing** – Continuous process improvement based on feedback and innovation.

One-line Viva Answer

CMM is a five-level framework that helps organizations evaluate and improve the maturity of their software development processes.

50. What is Regression Testing?

Testing existing functionality after changes to ensure nothing previously working has broken.

IoT

51. What is IoT?

IoT (Internet of Things) is a network of physical devices connected to the internet that collect, exchange, and act on data automatically.

52. What is ESP32?

ESP32 is a 32-bit microcontroller with built-in Wi-Fi and Bluetooth, widely used for IoT applications.

53. ESP32 vs Arduino Uno

ESP32	Arduino Uno
32-bit	8-bit
240 MHz	16 MHz
Wi-Fi & Bluetooth	None
More RAM	Less RAM

More GPIO	Fewer GPIO
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54. Why did you choose ESP32?

- Built-in Wi-Fi and Bluetooth
 - Faster processor
 - More GPIO pins
 - Supports ADC, PWM, I²C, SPI, and UART
 - Ideal for IoT projects
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55. What is a Microcontroller?

A microcontroller is a single chip containing a CPU, memory, and input/output peripherals used to control embedded systems.

56. What is a GPIO Pin?

GPIO (General Purpose Input/Output) is a programmable pin that can act as an input or output.

57. What is ADC?

ADC (Analog-to-Digital Converter) converts analog voltages into digital values that the microcontroller can process.

58. What is PWM?

PWM (Pulse Width Modulation) controls the average voltage by varying the duty cycle, commonly used for motors, LEDs, and servos.

59. Analog vs Digital Sensors

Analog sensors output continuous voltage values, while digital sensors output discrete digital signals.

60. What is a Pull-up Resistor?

A pull-up resistor keeps a digital input at a HIGH logic level when no signal is connected, preventing floating inputs.

61. What is I²C?

A two-wire communication protocol using SDA (data) and SCL (clock) that allows multiple devices to communicate with a controller.

62. What is SPI?

A high-speed synchronous communication protocol using MOSI, MISO, SCK, and CS lines, supporting full-duplex communication.

63. What is UART?

UART is an asynchronous serial communication protocol using TX and RX pins for point-to-point communication.

64. Difference between I²C, SPI, and UART

Protocol	Wires	Speed	Communication
I ² C	2	Medium	Multi-device bus
SPI	4+	High	Full-duplex
UART	2	Medium	Point-to-point

65. What is Baud Rate?

Baud rate is the number of signal changes (symbols) transmitted per second. In UART, it is commonly used to represent bits per second, such as 9600 or 115200.

66. What is the function of `pinMode()`?

It configures a GPIO pin as `INPUT`, `OUTPUT`, or `INPUT_PULLUP`.

67. `digitalRead()` VS `analogRead()`

- `digitalRead()` reads HIGH or LOW.
 - `analogRead()` reads a range of analog values (e.g., 0–4095 on the ESP32).
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68. `digitalWrite()` VS `analogWrite()`

- `digitalWrite()` outputs HIGH or LOW.
 - `analogWrite()` outputs a PWM signal to simulate analog output.
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69. Why do we use `setup()` and `loop()`?

- `setup()` runs once for initialization.
 - `loop()` runs continuously to execute the main program.
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70. What happens if an ultrasonic sensor detects no object?

No echo is received, so the function may return zero or a timeout value, indicating no object within range.

71. Why divide by 2 in the ultrasonic distance formula?

The ultrasonic pulse travels to the object and back, so dividing by 2 gives the one-way distance.

72. Why does a Servo require PWM?

A servo determines its angle from the width of PWM pulses, allowing precise position control.

73. DHT11 vs DHT22

DHT11	DHT22
Lower accuracy	Higher accuracy
0–50°C	–40–80°C
20–80% RH	0–100% RH
Cheaper	More expensive

74. Purpose of the Serial Monitor

It is used to display debug messages, sensor readings, and communicate with the microcontroller during development.

75. HTTP vs MQTT

- **HTTP:** Request–response protocol, heavier, suitable for web communication.
 - **MQTT:** Lightweight publish–subscribe protocol, efficient for IoT devices with limited bandwidth and power.
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Project-Specific IoT Viva Questions

76. Why did you choose ESP32 instead of Arduino Uno?

I chose the ESP32 because it provides built-in Wi-Fi and Bluetooth, a faster 32-bit processor, more memory, and more peripherals. These features make it much more suitable for IoT applications than the Arduino Uno.

77. Why did you use DHT22 instead of DHT11?

The DHT22 provides higher accuracy, a wider temperature and humidity measurement range, and better long-term reliability, making it suitable for my project.

78. How does the OLED display communicate with the ESP32?

The OLED uses the **I²C protocol**, where the ESP32 communicates through the SDA and SCL pins to send text and graphics.

79. What happens if the Wi-Fi connection is lost?

The ESP32 cannot send data to cloud services, but it can continue local sensor monitoring and display readings. The program can periodically attempt to reconnect to the Wi-Fi network.

80. How would you improve your project in the future?

I would add:

- Cloud storage and dashboard
 - Mobile app notifications
 - OTA (Over-the-Air) firmware updates
 - Battery backup
 - Data encryption
 - Machine learning for predictive analysis
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81. If deploying in a real smart home or agriculture system, what additional sensors or safety features would you add?

For agriculture:

- Soil moisture sensor
- Rain sensor
- Light intensity sensor
- pH sensor
- GPS module

For smart homes:

- Smoke and gas sensors
- Flame detector
- Motion sensor
- Door/window sensors
- Camera
- Battery backup
- Buzzer and emergency alerts

- Automatic power cut-off for hazardous conditions

These answers are concise enough for a viva while still demonstrating understanding. If the examiner asks follow-up questions, you can expand on the concepts with examples from your projects, especially your **MobileNetV3 mango classification thesis** and **ESP32-based IoT projects**.